# Subject 4: Prediction of the Critical Temperature of Superconducting Materials

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## 1 Introduction

This project studies the UCI superconductivity dataset. The statistical learning problem is a supervised regression problem: for each superconducting material, the target variable is the critical temperature $Y = \texttt{critical_temp}$, measured in Kelvin, and the covariates $X \in \mathbb{R}^{81}$ are physical and chemical descriptors of the material. The objective is to build a predictor $\widehat{f}(X)$ that estimates the critical temperature from these covariates.

## 2 Dataset Overview

The raw dataset contains 21,263 materials and 81 numerical explanatory variables. The version used in the code is fetched from the UCI repository with dataset id 464 and is cached locally by the reproducible Python pipeline. We found 66 exact duplicate rows in the full feature-target table. These duplicates were removed before the train-test split to avoid having identical observations in both sets. The processed modeling table therefore contains 21,197 observations and no missing values. The target is strongly right-skewed: its median is 20 K, while its mean is approximately 34.49 K and its maximum is 185 K.

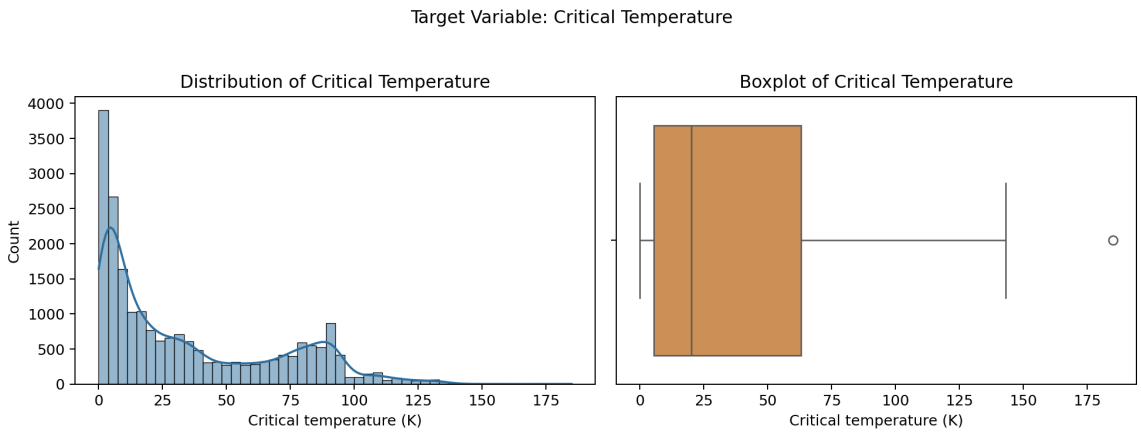


Figure 1: Distribution and boxplot of the critical temperature. The target is positive and right-skewed, with a long upper tail.

## 3 Exploratory Analysis

The exploratory analysis focuses on the target distribution, missing values, feature summaries, and marginal correlations between individual features and the target. Several weighted descrip-

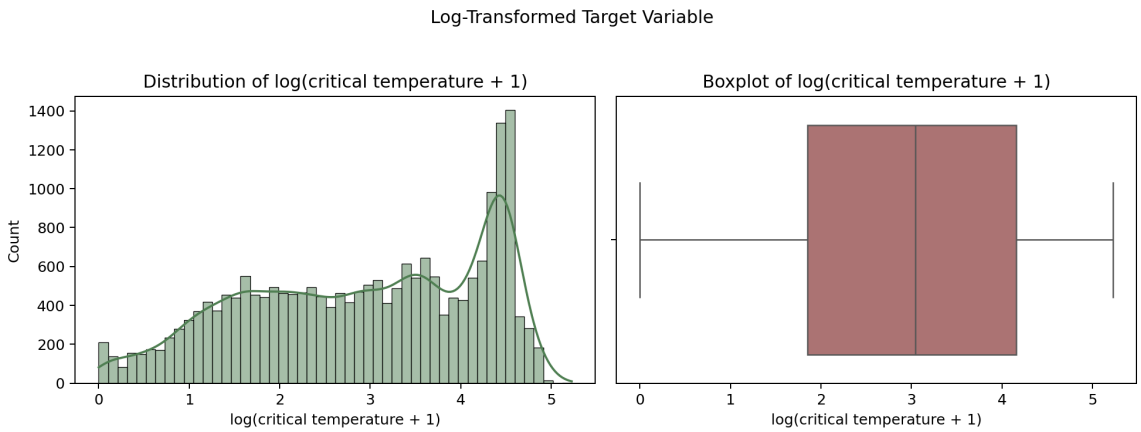


Figure 2: Distribution and boxplot after the transformation $\log(1 + \texttt{critical_temp})$. This transformation reduces the skewness of the target and is used inside the modeling pipelines.

tors of valence, atomic radius, electron affinity, and thermal conductivity are among the variables most correlated with the critical temperature.

# 4 Preprocessing

The preprocessing removes exact duplicate full rows, then splits the observations once into a training set and a held-out test set with an 80/20 split and random seed 42. Model selection and hyperparameter calibration are performed only on the training set. For OLS, Ridge, Lasso, and k-nearest neighbors, the covariates are standardized inside a scikit-learn pipeline so that scaling is fit only on the training folds. The target is transformed by $\log(1+Y)$ during training and predictions are transformed back with $\exp(\cdot) - 1$ before evaluation. All reported errors are therefore expressed on the original Kelvin scale.

# 5 Methods

The baseline model predicts a constant average target value. The course-covered linear methods are ordinary least squares, Ridge regression, and Lasso regression. Ridge and Lasso use cross-validation on the training set to calibrate their regularization parameters. We also include k-nearest-neighbor regression, with the number of neighbors selected by cross-validation.

As an additional nonlinear benchmark, we use a histogram-based gradient boosting regressor. This method is not treated as a course-covered method in the report. It builds an additive model of shallow regression trees, where each new tree is fit to reduce the current prediction error. It is useful here because the relation between the physical descriptors and the critical temperature is unlikely to be purely linear.

# 6 Evaluation Criteria

For observations $(y_i, \widehat{y}_i)_{i=1}^n$ in the held-out test set, the main criteria are

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}},$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|, \qquad R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

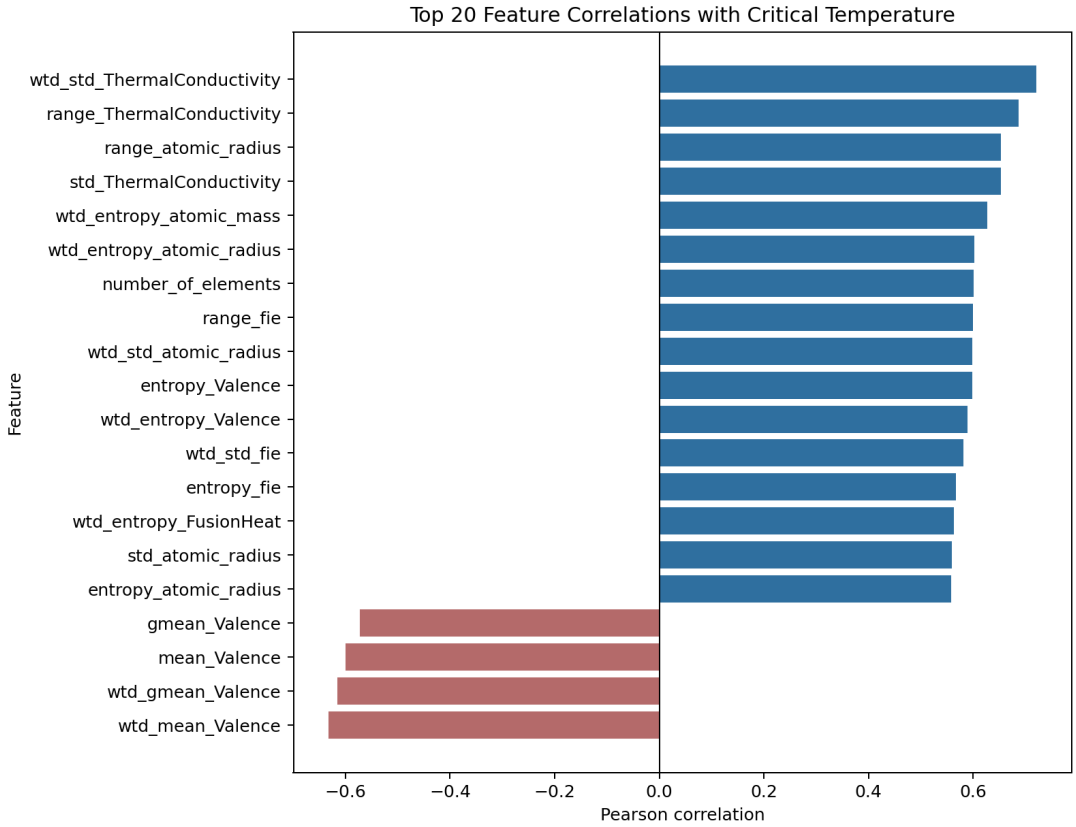


Figure 3: The twenty features with the largest absolute Pearson correlation with the target. These correlations are descriptive only and do not by themselves define a final model.

RMSE is used as the primary selection criterion because it is expressed in Kelvin and penalizes large errors strongly. MAE and $R^2$ are used as complementary criteria.

# 7 Results

| Model | MSE | RMSE | MAE | R2 |
|---|---|---|---|---|
| knn_gridsearch | 113.5725 | 10.6570 | 5.6747 | 0.9035 |
| hist_gradient_boosting | 115.4570 | 10.7451 | 6.2916 | 0.9019 |
| ridge_cv | 334.8801 | 18.2997 | 12.4752 | 0.7155 |
| linear_regression | 334.9093 | 18.3005 | 12.4742 | 0.7155 |
| lasso_cv | 336.1832 | 18.3353 | 12.5209 | 0.7144 |
| dummy_regressor | 1469.0342 | 38.3280 | 27.6689 | -0.2480 |

Table 1: Held-out test performance. The comparison uses only the test set, after model selection on the training data.

The nonlinear models outperform the linear family by a large margin. The k-nearest-neighbor model obtains the lowest RMSE, approximately 10.66 K, the lowest MAE, approximately 5.67 K, and the highest $R^2$, approximately 0.904. The histogram-based gradient boosting model is very close, with RMSE approximately 10.75 K and $R^2$ approximately 0.902. The two nonlinear methods are therefore close in predictive performance, while OLS, Ridge, and Lasso remain around RMSE 18.3 K. This gap suggests that the relation between the physical descriptors

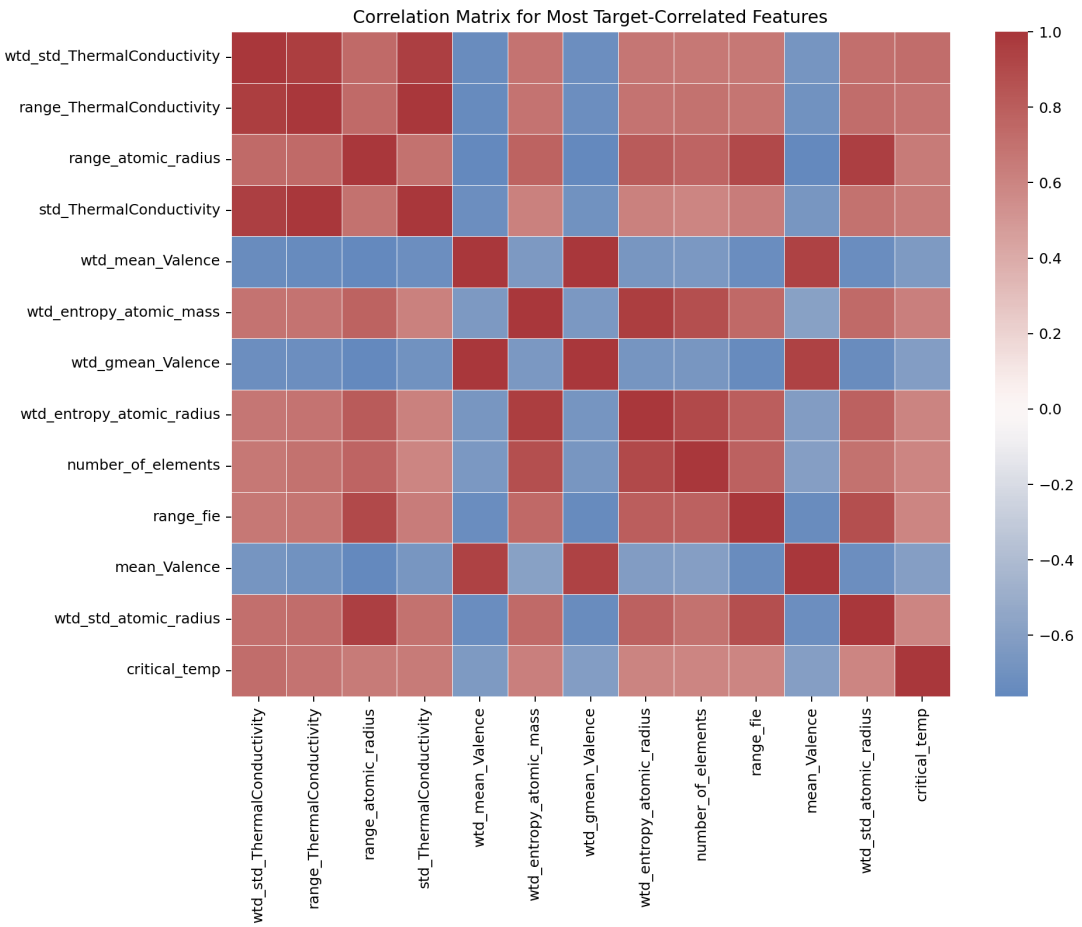


Figure 4: Correlation matrix for the features most correlated with the target. The visible dependence between covariates motivates regularized models such as Ridge and Lasso.

and critical temperature contains nonlinear structure or interactions that linear models do not capture well.

## 8 Limitations

The final nonlinear model is less directly interpretable than the regularized linear models. The analysis also uses the engineered UCI features rather than reconstructing descriptors from chemical formulae. Finally, the target transformation improves numerical behavior but means that the fitted objective is not exactly the same as directly minimizing squared error in Kelvin.

## 9 Conclusion

Based on the held-out test RMSE, MAE, and $R^2$, the recommended model is k-nearest-neighbor regression with $k = 3$ selected by cross-validation. It substantially improves predictive accuracy compared with OLS, Ridge, and Lasso. The histogram-based gradient boosting regressor is a very close nonlinear alternative and has the practical advantage of producing a compact fitted model that does not require storing all training observations for prediction. If interpretability is prioritized over predictive performance, Ridge regression is the most stable linear alternative, but it loses a large amount of accuracy on this dataset.

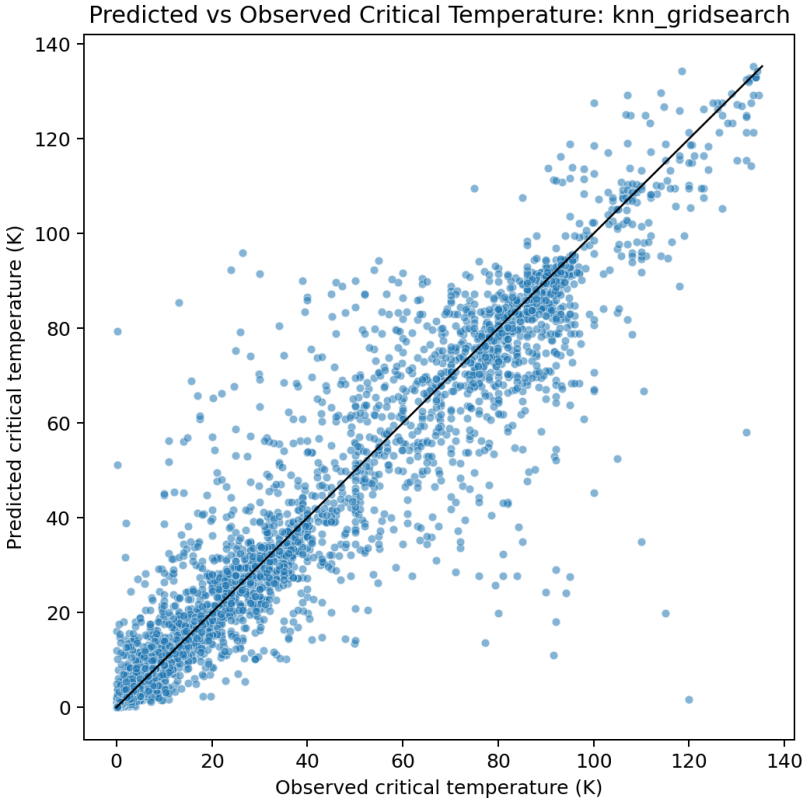


Figure 5: Predicted versus observed critical temperature for the selected model. Points near the diagonal correspond to accurate predictions on the Kelvin scale.

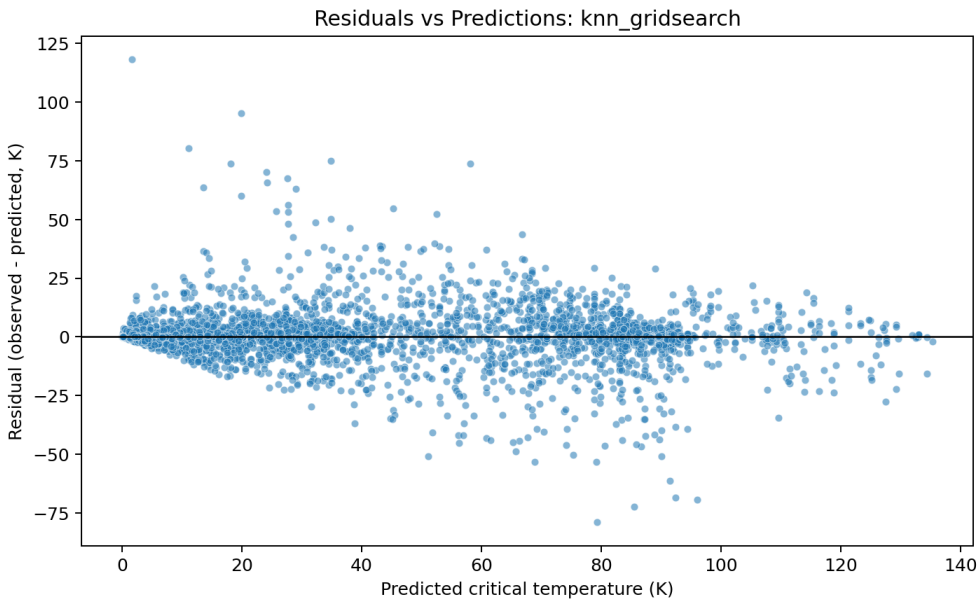


Figure 6: Residuals of the selected model as a function of predicted critical temperature. This diagnostic checks whether large errors are concentrated in specific prediction ranges.